INCH-POUND

M1L-L-85776A(AS)
30 November 1990
SUPERSEDING
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8 November 1985

MILITARY SPECIFICATION

# LAUNCHER, ROCKET LMU-24A/E AND LMU-24B/E

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification establishes the requirements for the manufacture and inspection of the Launcher, Rocket LMU-24A/E and LMU-Z4B1E.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection of Attributes |
| MIL-STD-109 | Quality Assurance Terms and Definitions |
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-81O | Environmental Test Methods and Engineering Guidelines |
| MIL-STD-2073-1 | DOD Materiel Procedures for Development and Application of Packaging Requirements |
| DOD-STD-21O1 | Classification of Characteristics |
| MIL-STD-45662 | Calibration System Requirements |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use In improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 53, Lakehurst, NJ 08733-5100, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at

AMSC N/A FSC 1055

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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.1.2 Other Government documents. drawings and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAwINGS

NAVAL AIR SYSTEMS COMMAND (CAGE Code 30003)

| | |
|---|---|
| DL1335AS700 | Launcher, Rocket LMU-24A/E |
| DL1335AS900 | Launcher, Rocket LMU-24B/E |
| 1335AS260 | Igniter, PVU-3B/E |
| 1335AS275 | Igniter, PVU-3A/E |

(Application for copies should be addressed to Commanding Officer, Naval Air Technical Services Facility (NATSF) (Code 3121), 700 Robbins Avenue, Philadelphia, PA 19111-5096).

2.2 Non-Government Duplications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issue of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issue of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

| | |
|---|---|
| ANSI B46.1 | Surface Texture (Surface Roughness Waviness and Lay |

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 1OO18-33O8.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing In this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

3.1 First article. when specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Material and processes. The material and processes utilized in the fabrication of the launcher shall conform to the requirements indicated on the drawings and specifications listed on DL1335AS700 and DL1335AS900 (see 4.1.3, 6.2, and 6.3).



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3.2.1 Recycled virgin and reclaimed materials. There is no exclusion to the use of recycled or reclaimed materials and no mandate for the use of virgin materials provided it meets the requirements of this specification.

3.3 Design and construction. The launcher shall be of design, construction, and physical dimensions specified on the drawings and specifications listed on DL1335AS700 and DL1335AS900.

3.4 Environmental conditioning. The launcher shall be capable of exposure to high temperature, low temperature, and humidity without degradation of performance (see 4.6.2, 4.6.3, and 4.6.4).

3.5 Performance.

3.5.1 Recycle time and amperage (M101). When operated with a fully charged battery, the time required for monitared voltage to regain a level of 195 ± 20 volts direct current (Vdc) for subsequent actuation shall be not greater than 360 seconds. The amperage from the battery, at the end of this time, shall be not greater than 140 milliamperes (see 4.6.5).

3.5.2 Firing pin penetration (M102). When actuated, the firing pin shall produce a 0.015 inch minimum deep depression without rupture in the percussion primer of the PVU-3B/E igniter, Drawing 1335AS260, or the PVU-3A/E igniter, Drawing 1335AS275 (see 4.6.6).

3.5.3 Discharge time and voltage test (M103) After removal of the enable plug, the time for the voltage drop from 195 ± 20 volts to not greater than 50 volts shall be not greater than 4 seconds (see 4.6.7).

3.6 Workmanship. The workmanship requirements are as specified on the drawings and the following. Workmanship and finish shall be in accordance with the highest grade practice used in manufacturing military weapons. Finished items and parts shall not exhibit poor material and processing such as seams, laps, laminations, cracks, visible steps, sharp edges, nicks, scratches, burrs, deformations, and missing operations which may affect serviceability, functioning, operation, appearance, or safety. Fins and other extraneous metal shall be removed from cast or forged parts. Hammering to shape, salvage operations (including repair by welding), or other similar practices shall not be permitted without prior approval of the contracting officer. No part or assembly shall contain chips, dirt, grease, rust, corrosion, or any other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection Unless otherwise specified in the contract or purchase order, the contractor Is responsible for, the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements (see 6.2).



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4.1.1 Responsibi]ity for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Test requirement and inspection facilities. The manufacturer shall ensure that test and inspection facilities of sufficient accuracy, quality, and quantity are established and maintained to permit performance of required inspections. All test and measurement equipment shall be calibrated and maintained in accordance with MIL-STD-45662. Unless otherwise specified herein, all test and measurement equipment shall have an accuracy within one-fifth of the tolerance for the variable to be measured.

4.1.3 Certification. When specified, a written certification (see 6.2 and 6.3), accompanied by objective quality evidence (as defined in MIL-STD-109) that the materials and processes conform to the requirements Indicated on the drawings and specifications listed on DL1335AS700 and DL1335AS900, shall be available to the acquiring activity.

4.1.4 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as major or minor in accordance with DOD-STD-21O1. Requirements that are major characteristics are identified by the symbol (M). The number following the classification symbol indicates the serial number of the requirement. Requirements which are not annotated with a classification code are classified as minor.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article .inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "General Requirements" of MIL-STD-81O (see 6.2).

4.4 First article lnspection. First article inspection shall be performed *on* sample units produced with equipment and procedures normally used in production.

4.4.1 Sample Unless otherwise specified, first article sample shall consist of five launchers and five sets of unassembled metal parts (see 6.2 and 6.4).

4.4.2 Inspection routine. The first article inspection shall consist of the inspections specified in table I. Failure to pass any one of the inspections shall be cause for rejection.



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TABLE I. First article inspection.

| Inspection | Requirement paragraph | Method paragraph | Number of samples |
|---|---|---|---|
| Visual | 3.3 | 4.6.1 | 10 1/ |
| Environmental | | | |
| High temperature | 3.4 | 4.6.2 | 5 2/ |
| Low temperature | 3.4 | 4.6.3 | 5 2/ |
| Humidity | 3.4 | 4.6.4 | 5 2/ |
| Performance | | | |
| Recycle time and amperage (M101) | 3.5.1 | 4.6.5 | 5 2/ |
| Firing pin penetration (M102) | 3.5.2 | 4.6.6 | 5 2/ |
| Discharge time and voltage (M103) | 3.5.3 | 4.6.7 | 5 2/ |

1/ Samples consist of five launchers and five sets of unassembled metal parts.
2/ Samples consist of five launchers.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of Group A inspection (see 6.6.1) specified in table II.

TABLE II. Group A inspection.

| Inspection | Requirement paragraph | Method paragraph | Sampling procedures |
|---|---|---|---|
| Visual | 3.2, 3.6 | 4.6.1 | 4.5.2 |
| Performance | | | |
| Recycle time and amperage (M1OI) | 3.5.1 | 4.6.5 | 4.5.3 |
| Firing pin penetration (M102) | 3.5.2 | 4.6.6 | 4.5.3 |
| Discharge time and voltage (M103) | 3.5.3 | 4.6.7 | 4.5.3 |

4.5.1 Inspection lot. An inspection lot shall be as defined in MIL-STD-105 and shall consist of all launchers offered for inspection during a period of 1 month provided the homogeneity requirements of inspection lots have-been retained.

4.5.2 Sampling plan. The sample size for visual inspection shall be in accordance with MIL-STD-105, inspection level II.

4.5.3 Individual inspection. Individual inspection is inspection conducted on each launcher (100 percent inspection). Failure to pass inspection shall be cause for rejection of the launcher.

4.6 Methods of inspection.

4.6.1 Visual examination. Visual examination shall be made for such defects as defective metal, burrs, sharp corners, foreign matter, protective finish defects, marking missing, incorrect or illegible, thread damaged or not full.



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holes, slots, radii or chambers missing. Four power (4x) magnification may be used in performing visual examination for such defects as burrs and foreign matter. The roughness comparison specimens prescribed in ANSI B46.1 shall be use as a basis of comparison for surface roughness (finish) determination.

4.6.2 High temperature operation. The launcher shall be placed In a conditioning chamber at 160 ± 5°F for a period of not less than 4 hours and not greater than 16 hours. The battery shall be removed from the electronics enclosure before performing this test.

4.6.3 Low temperature operation. The launcher shall be placed in a conditioning chamber at -40 ± 5°F for a period of not less than 4 hours and not greater than 16 hours. The battery shall be removed from the electronics enclosure before performing this test.

4.6.4 Temperature and humidity. A temperature and humidity test shall be performed on the launcher in accordance with MIL-STD-81O, Method 507.3, Procedure III. The duration of the test shall be four 24-hour cycles.

4.6.5 Recycle time and amperage (M101). The battery shall be fully charged for the recycle test. The voltage from the J5-A, J6-A, J7-A, and J8-A connectors to the negative side of the battery shall be monitored using a multimeter with an input impedance of not less than 20,000 ohms per volt (ohms/volt). The firing switch shall then be activated for each position and the amperage from the battery shall be monitored using a current meter with an accuracy of not less than 2 percent of full scale.

4.6.6 Firing Pin penetration (M102). A PVU-3B/E igniter, Drawing 1335AS260 -or a PVU-3A/E.igniter, Drawing 1335AS275, devoid of all explosive materials except the M42 primer, shall be loaded into the launcher breech. The firing switch shall be actuated after a capacitor charging interval of 140 ± 10 seconds.

4.6.7 Discharge time and voltage (M103) The battery shall be fully charged for the discharge test. The voltage from the J5-A, J6- A, J7-A, and J8-A connectors to the negative side of the battery shall be monitored using a multimeter with an impedance of not less than 20,000 ohms/volt. The enable plug shall be removed after the system has stabilized at 195 ± 20 volts.

4.7 Inspection of Packaging. The sampling and inspection of the preservation, packing, and container marking shall conform to this specification (see section 5).

## 5. PACKAGING

5.1 Preservation. Unless otherwise specified, preservation shall be in accordance with MIL-STD-2073-1, level C (see 6.2).

5.2 Packing. Unless otherwise specified, packing shall be in accordance with MIL-STD-2073-1, level C (see 6.2).

5.3 Marking. In addition to any special marking or other identification required by the contract or purchase order (see 6.2), unit packages and shipping containers shall be marked in accordance with MIL-STD-129.



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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The LMU-24A/E and the LMU-24B/E launchers are intended for use with the GTR-18A rocket and the PVU-3B/E or the PVU-3A/E igniter to train aviation and ground forces to recognize and cope with attacking surface-launch missiles.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Part number.
d. Quantity required.
e. Whether first article is required (see 3.1, 4.4 and 6.4).
f. Name and location of Government approved test laboratory.
g. Inspection conditions, if other than as specified in 4.3.
h. First article sample, if other than as specified in 4.4.1.
i. Level of preservation and packing, if other than as specified in 5.1 and 5.2.
j. Marking if other than as specified in 5.3.
k. Where data items (see 6.3) are to be forwarded.

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (I)D Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

| Reference Paragraph | DID Number | DID Title | Suggested Tailoring |
|---|---|---|---|
| 4.1.3 | DI-E-2121 | Certificate of compliance | --- |

The above DID's were those cleared as of the date of this specification. The current issue of DOD 501O.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of five launchers and five sets of unassembled metal parts. A first sample should be submitted whenever a manufacturer produces the launchers for the first time, whenever a major change in processing occurs, and whenever there is a lapse in production of 1 year or more. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles.



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6.5 quality assurance definitions. Definitions for quality assurance terms are in accordance with MIL-STD-109.

6.6 Definitions.

6.6.1 Group A inspection. Group A inspection consists of nondestructive inspections of all items produced or all samples from an inspection lot to demonstrate product compliance with contractual requirements. Group A inspections examine characteristics most affected by variations in production processes or skills and functions vital to successful completion of the design mission.

6.7 Subject term (key word) listing.

Electronics enclosure
Firing switch
Igniter, PVU-3A/E
Igniter, PVU-3B/E

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - AS
(Project 1055-N075)

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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

## INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER<br>MIL-L-85776A(AS) | 2. DOCUMENT DATE (YYMMDD)<br>30 November 1990 |
|---|---|---|

3. DOCUMENT TITLE
LAUNCHER, ROCKET LMU-24A/E and LMU-24B/E

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON (If applicable) | 7. DATE SUBMITTED (YYMMDD) |

8. PREPARING ACTIVITY

| a. NAME<br>Commanding Officer<br>NAEC (SESD) Code 53 | b. TELEPHONE (Include Area Code)<br>(1) Commercial<br>908/323-1116<br>(2) AUTOVON<br>624-1116 |
|---|---|
| c. ADDRESS (Include Zip Code)<br>Lakehurst, NJ 08733-5100 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 |

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